

Agenda

- Outliers
- Quartiles, percentiles
- Categorical data
- Plots

Outliers

- **Outliers** are points that are much larger or smaller than the rest of the sample points.
- Outliers may be data entry errors or they may be points that really are different from the rest.
- Outliers should not be deleted without considerable thought—sometimes calculations and analyses will be done with and without outliers and then compared.

Quartiles

Quartiles divide the data as nearly as possible into quarters.

The **first quartile** is the median of the lower half of the data.

- To find the first quartile, compute $0.25(n + 1)$. If this is an integer, then the sample value in that position is the first quartile. If not, take the average of the sample values on either side of this value.

Quartiles

The **third quartile** is the median of the upper half of the data.

- To find the third quartile, compute $0.75(n + 1)$. If this is an integer, then the sample value in that position is the third quartile. If not, take the average of the sample values on either side of this value.

Note: The computation we used for the location of the median is equivalent to $0.5(n + 1)$. The median is the second quartile.

Example

Suppose we have the following data:

2, 3, 5, 6, 7, 9, 9, 11, 12, 15

- What is the mean of these data?
- What is the median?
- What is the first quartile?
- What is the third quartile?

Definition of Percentile

- The ***p*th percentile** of a sample, for a number p between 0 and 100, divides the sample so that as nearly as possible $p\%$ of the sample values are less than the p th percentile, and $(100 - p\%)$ are greater.
- The computation of the location of the p th percentile is analogous to what we did for the quartiles.

To Find Percentiles

- Order the n sample values from smallest to largest.
- Compute the quantity $\left(\frac{p}{100}\right)(n + 1)$, where n is the sample size.
- If this quantity is an integer, the sample value in this position is the p th percentile. Otherwise, average the two sample values on either side.

Note on Percentiles

- The **first quartile** is the 25th percentile.
- The **median** is the 50th percentile.
- The **third quartile** is the 75th percentile.

Summary Statistics for Categorical Data

The two most commonly used numerical summaries for categorical data are the **frequencies** and the **sample proportions** (sometimes called **relative frequencies**).

- The frequency for a given category is simply the number of sample items that fall into that category.
- The sample proportion is the frequency divided by the sample size.

Summary Statistics for Categorical Data

- Example: 100 rivets are checked for their breaking strength. If 4 of the rivets fail (that is: do not hold up to the standard), find the sample proportion of rivets that fail.

Sample Statistics and Population Parameters

- Each sample statistic in this chapter has a population parameter counterpart.
- Numerical summaries of a sample are called **statistics**.
- Numerical summaries of a population are called **parameters**.
- Statistics are often used to estimate parameters.

Graphical Summaries

- Stem-and-leaf plot.
- Dot plot.
- Histogram.
- Boxplot.
- Scatterplot.

Stem-and-Leaf Plot

- A simple way to summarize a data set.
- Each item in the sample is divided into two parts: a **stem**, consisting of the leftmost one or two digits, and the **leaf**, which consists of the next digit.
- It is a compact way to represent the data.
- It also gives us some indication of the shape of our data.

Example 5

- Example: Duration of dormant periods of the geyser Old Faithful in Minutes.
- Stem-and-leaf plot:

Stem	Leaf
4	259
5	0111133556678
6	067789
7	01233455556666699
8	000012223344456668
9	013

- Let's look at the first line of the stem-and-leaf plot. This represents measurements of 42, 45, and 49 minutes.
- A good feature of these plots is that they display all the sample values. One can reconstruct the data in its entirety from a stem-and-leaf plot, although some digits may be truncated.

Stem-and-Leaf Diagrams

Compressive Strength (in psi) of aluminum-lithium alloy specimens

105	221	183	186	121	181	180	143
97	154	153	174	120	168	167	141
245	228	174	199	181	158	176	110
163	131	154	115	160	208	158	133
207	180	190	193	194	133	156	123
134	178	76	167	184	135	229	146
218	157	101	171	165	172	158	169
199	151	142	163	145	171	148	158
160	175	149	87	160	237	150	135
196	201	200	176	150	170	118	149

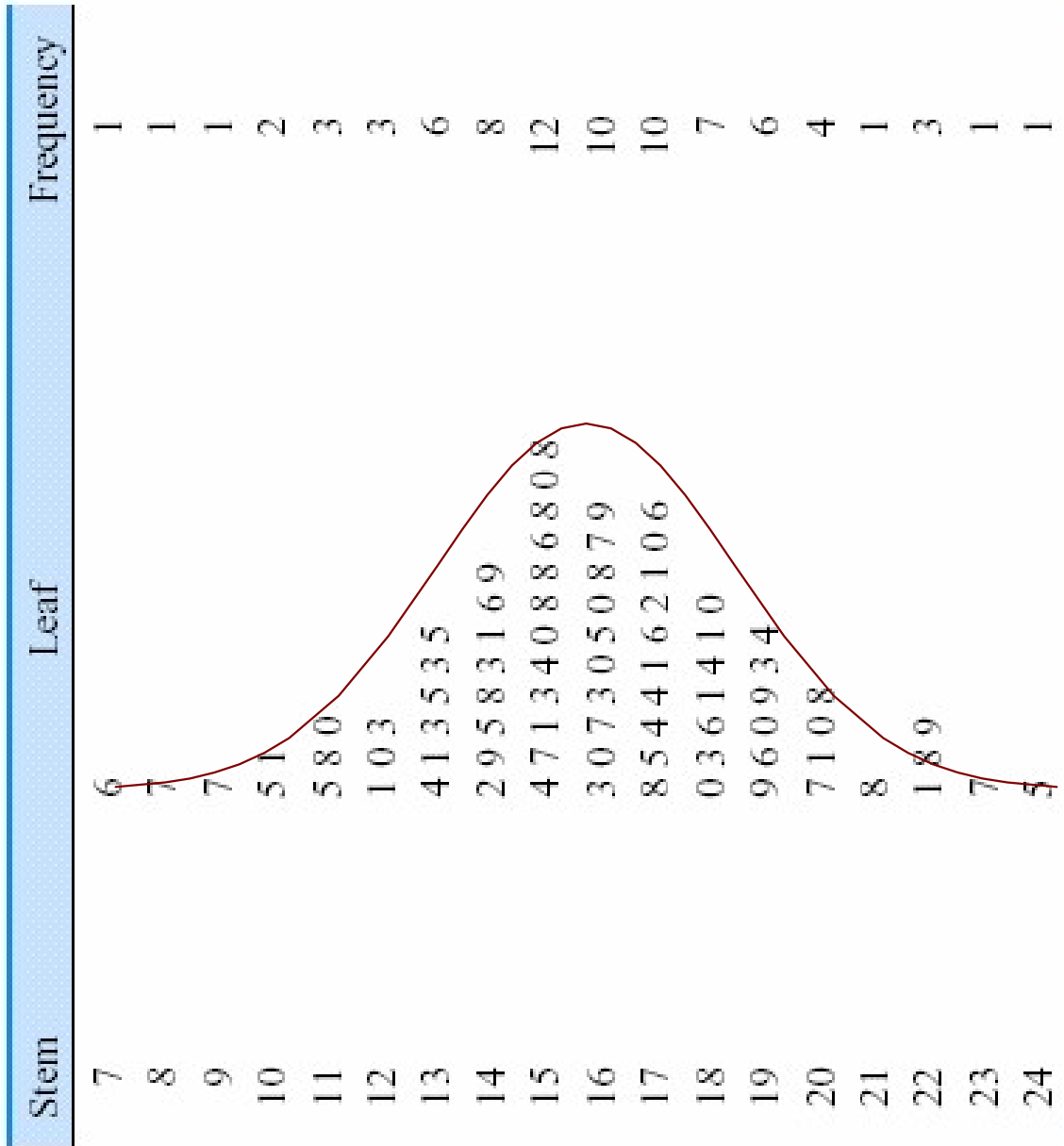
Stem-and-Leaf Diagrams

Stem	Leaf	Frequency
7	6	1
8	7	1
9	7	1
10	5 1	2
11	5 8 0	3
12	1 0 3	3
13	4 1 3 5 3 5	6
14	2 9 5 8 3 1 6 9	8
15	4 7 1 3 4 0 8 8 6 8 0 8	12
16	3 0 7 3 0 5 0 8 7 9	10
17	8 5 4 4 1 6 2 1 0 6	10
18	0 3 6 1 4 1 0	7
19	9 6 0 9 3 4	6
20	7 1 0 8	4
21	8	1
22	1 8 9	3
23	7	1
24	5	1

Stem-and-leaf diagram
for the compressive
strength data

Stem : Tens and hundreds digits (psi); Leaf: Ones digits (psi)

Stem-and-Leaf Diagrams

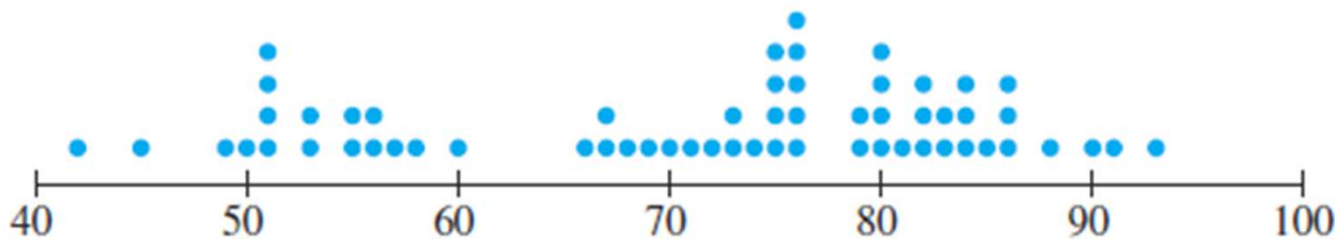


Stem : Tens and hundreds digits (psi); Leaf: Ones digits (psi)

Stem-and-leaf diagram
for the compressive
strength data

Dotplot

- A **dotplot** is a graph that can be used to give a rough impression of the shape of a sample, giving a good indication of where the sample values are concentrated and where the gaps are.
- It is useful when the sample size is not too large and when the sample contains some repeated values.
- It is a good method, along with the stem-and-leaf plot, to informally examine a sample. However, they are rarely used in formal presentations.
- Example (geyser data):



Histogram

- Graphical display that gives an idea of the “shape” of the sample, indicating regions where sample points are concentrated and regions where they are sparse.
- The bars of the histogram touch each other. A space indicates that there are no observations in that interval.

Creating a Histogram

Choose boundary points for the class intervals. Usually these intervals are the same width.

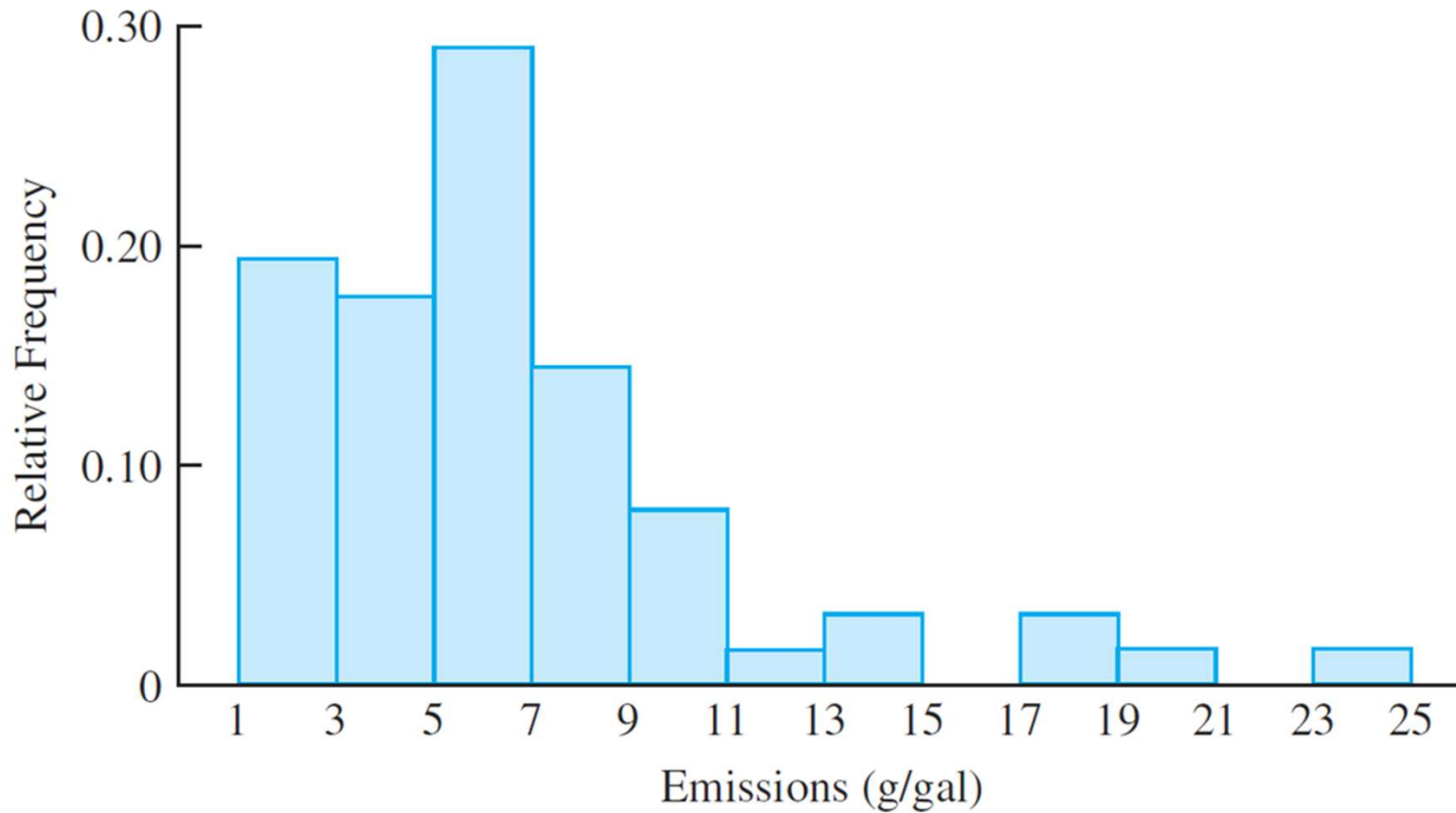
Compute the frequencies: this is the number of observations that occur in each interval.

Compute the relative frequencies for each class: this is the number of observations in each interval divided by the total number of observations.

If the class intervals are the same width, then draw a rectangle for each class, whose height is equal to the frequencies or relative frequencies.

- If the class intervals are of unequal widths, the heights of the rectangles must be set equal to the densities, where **density** is the relative frequency divided by the class width.

Example of Histogram



Symmetry and Skewness ¹

A histogram is perfectly **symmetric** if its right half is a mirror image of its left half.

- For example, the distribution of birth weights of full-term infants is roughly symmetric.

Histograms that are not symmetric are referred to as **skewed**.

A histogram with a long right-hand tail is said to be **skewed to the right**, or **positively skewed**.

- For example, the distribution of household incomes is right-skewed.

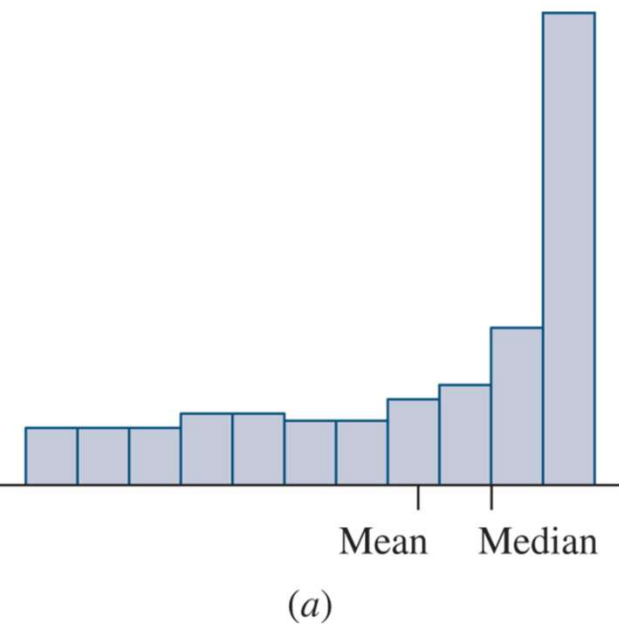
A histogram with a long left-hand tail is said to be **skewed to the left**, or **negatively skewed**.

- For example, the distribution of grades on an easy test is left-skewed.

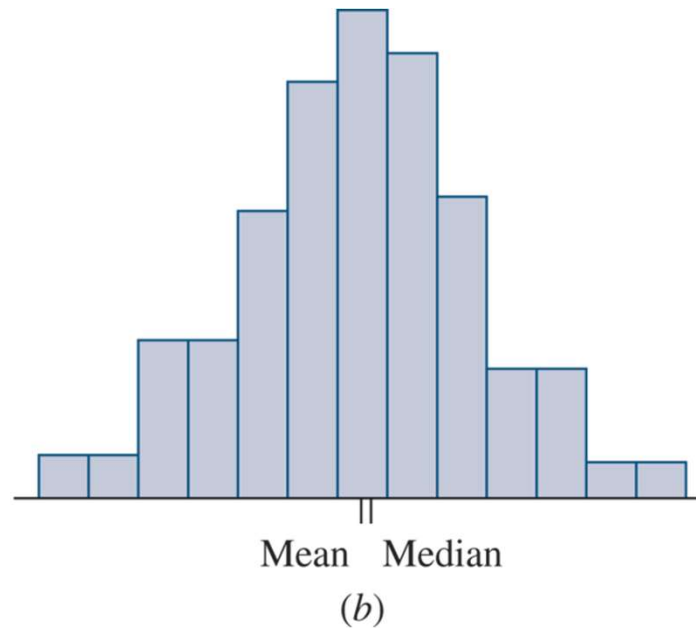
Symmetry and Skewness ²

- When a histogram is roughly **symmetric**, the mean and the median are approximately equal.
- When a histogram is **right-skewed**, the mean is greater than the median.
- When a histogram is **left-skewed**, the mean is less than the median.

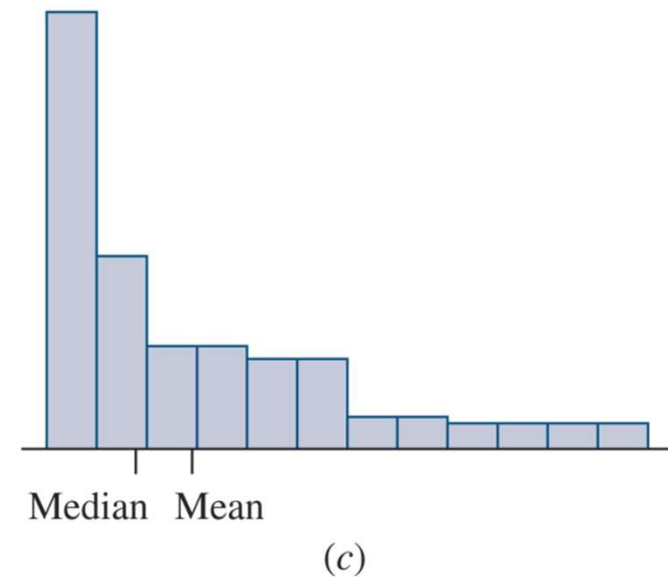
Skewness



Skewness < 0
Left skewed
(left tail is long)



Skewness ≈ 0



Skewness > 0
Right skewed
(right tail is long)

Unimodal and Bimodal

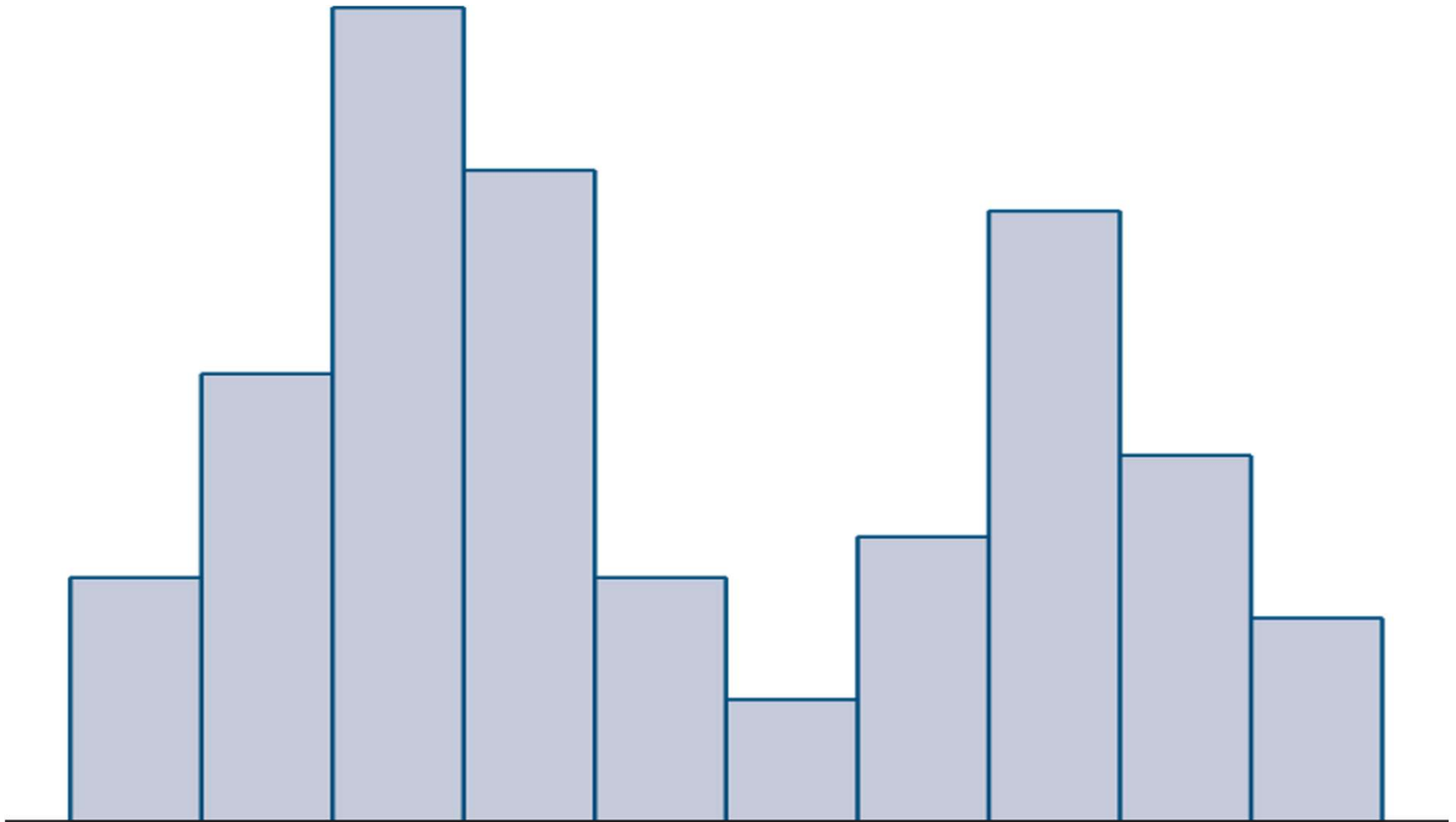
A histogram with only one peak is what we call **unimodal**.
If a histogram has two peaks then we say that it is **bimodal**.

- Bimodal histograms often indicate subsamples.

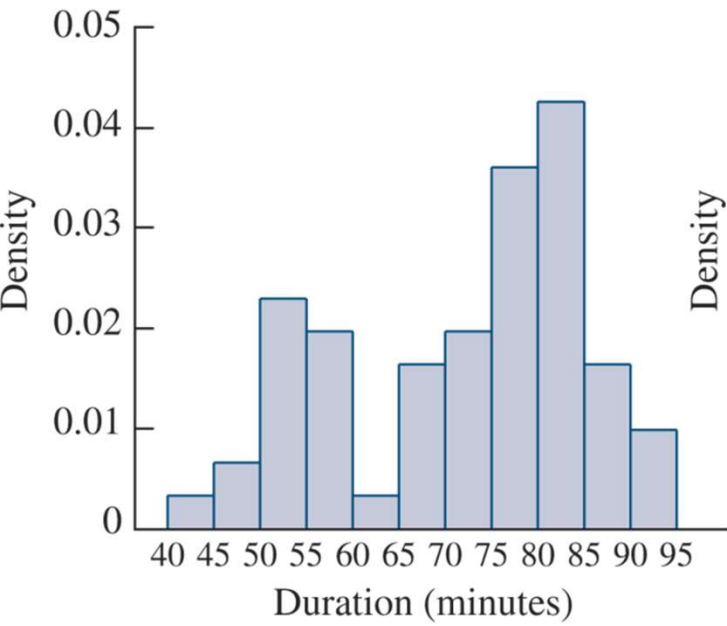
If there are more than two peaks in a histogram, then it is said to be **multimodal**.

- Multimodal histograms do not often occur in practice.

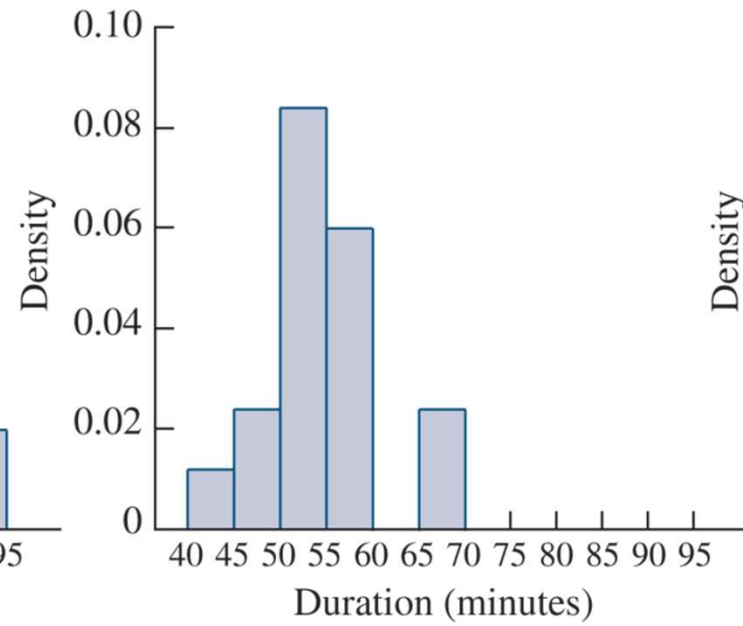
Bimodal data



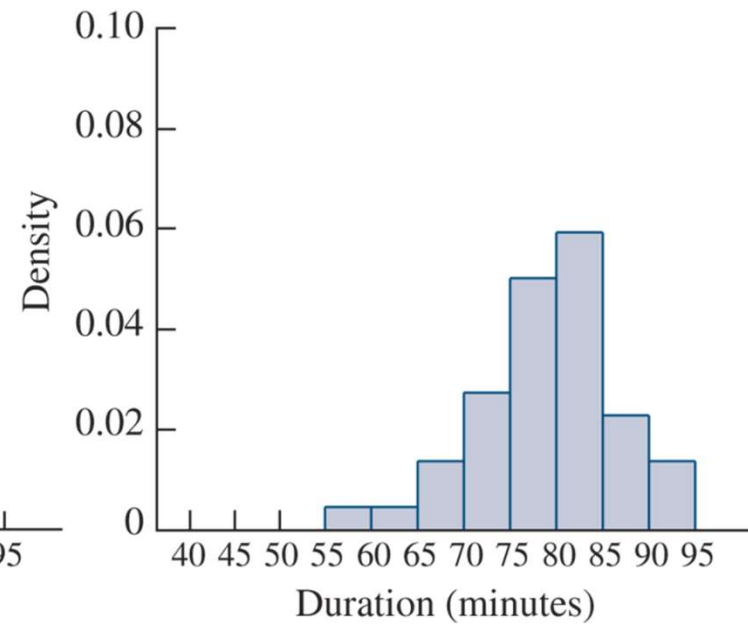
Example



(a)



(b)



(c)

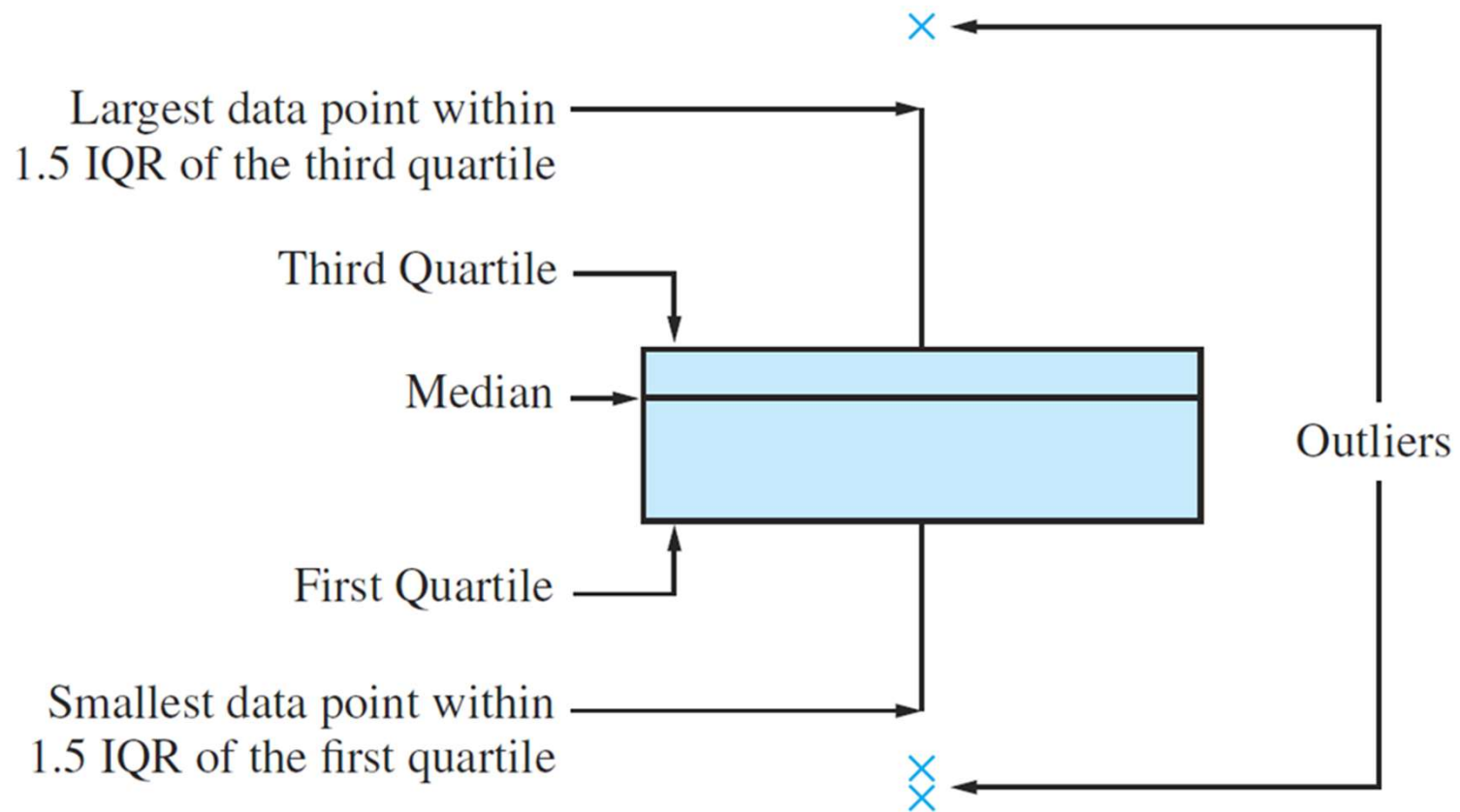
Boxplots

- A **boxplot** is a graphic that presents the median, the first and third quartiles, and any outliers present in the sample.
- The **interquartile range (IQR)** is the difference between the third quartile and the first quartile. This is the distance needed to span the middle half of the data.

Creating a Boxplot

- Compute the median and the first and third quartiles of the sample. Indicate these with horizontal lines. Draw vertical lines to complete the box.
- Find the largest sample value that is no more than 1.5 IQR above the third quartile, and the smallest sample value that is no more than 1.5 IQR below the first quartile. Extend vertical lines (whiskers) from the quartile lines to these points.
- Points more than 1.5 IQR above the third quartile, or more than 1.5 IQR below the first quartile, are designated as outliers. Plot each outlier individually.

Anatomy of a Boxplot



Agenda

- Tour of LearnHub
 - <https://learn.hub.ku.edu.tr/course/view.php?id=154#section-2>
 - [Course: INDR252-Fall25-LABA-CE-APPLIED STATISTICS | Learn](#)
- Class exercise
- Boxplots
- Propagation of error

Class exercise

How can you explain the following statement using the notion of sampling?

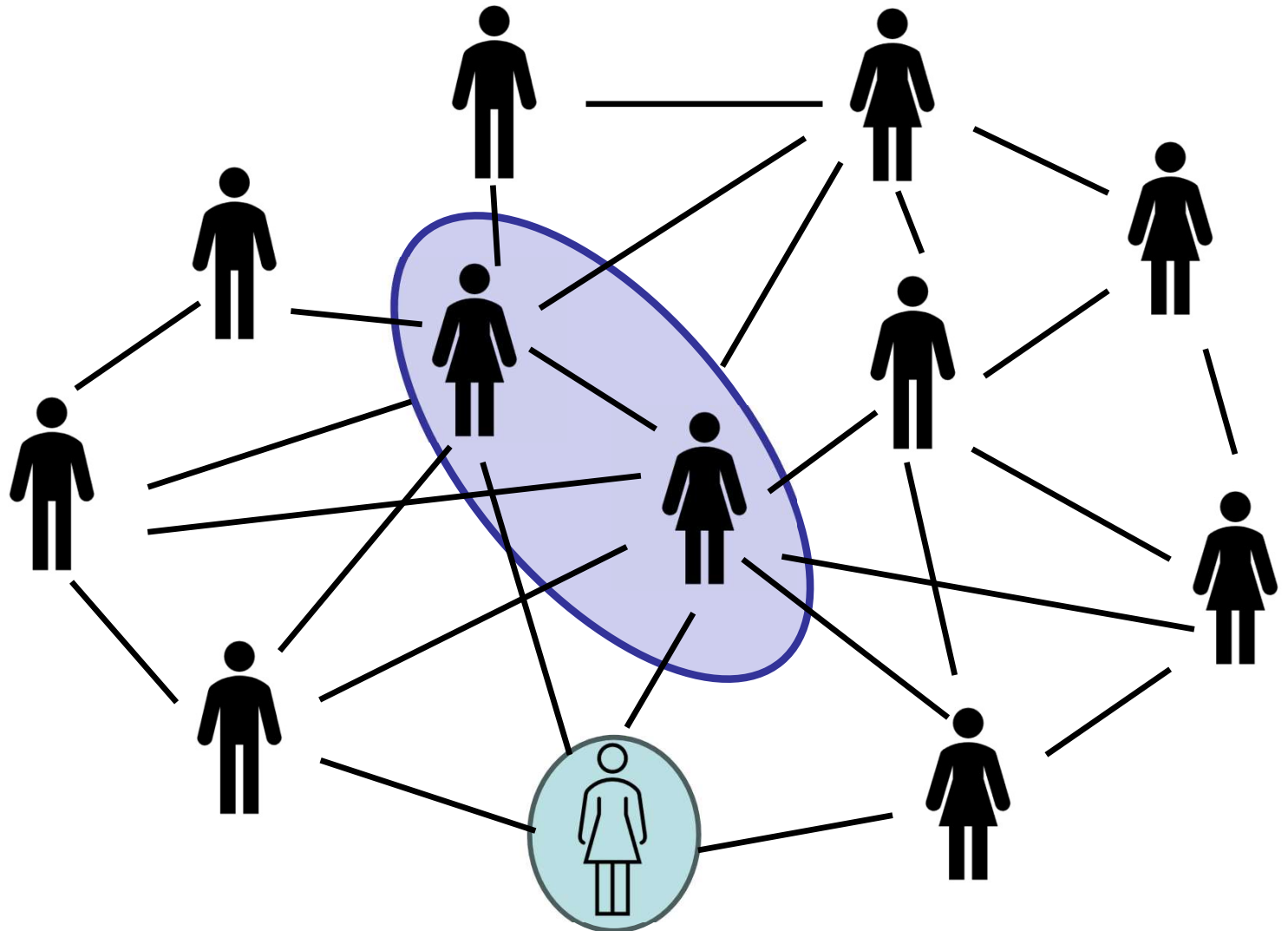
“On average, your friends have more friends than you do.”

What kind of assumptions (on the population and on the sample) are needed to justify this statement?

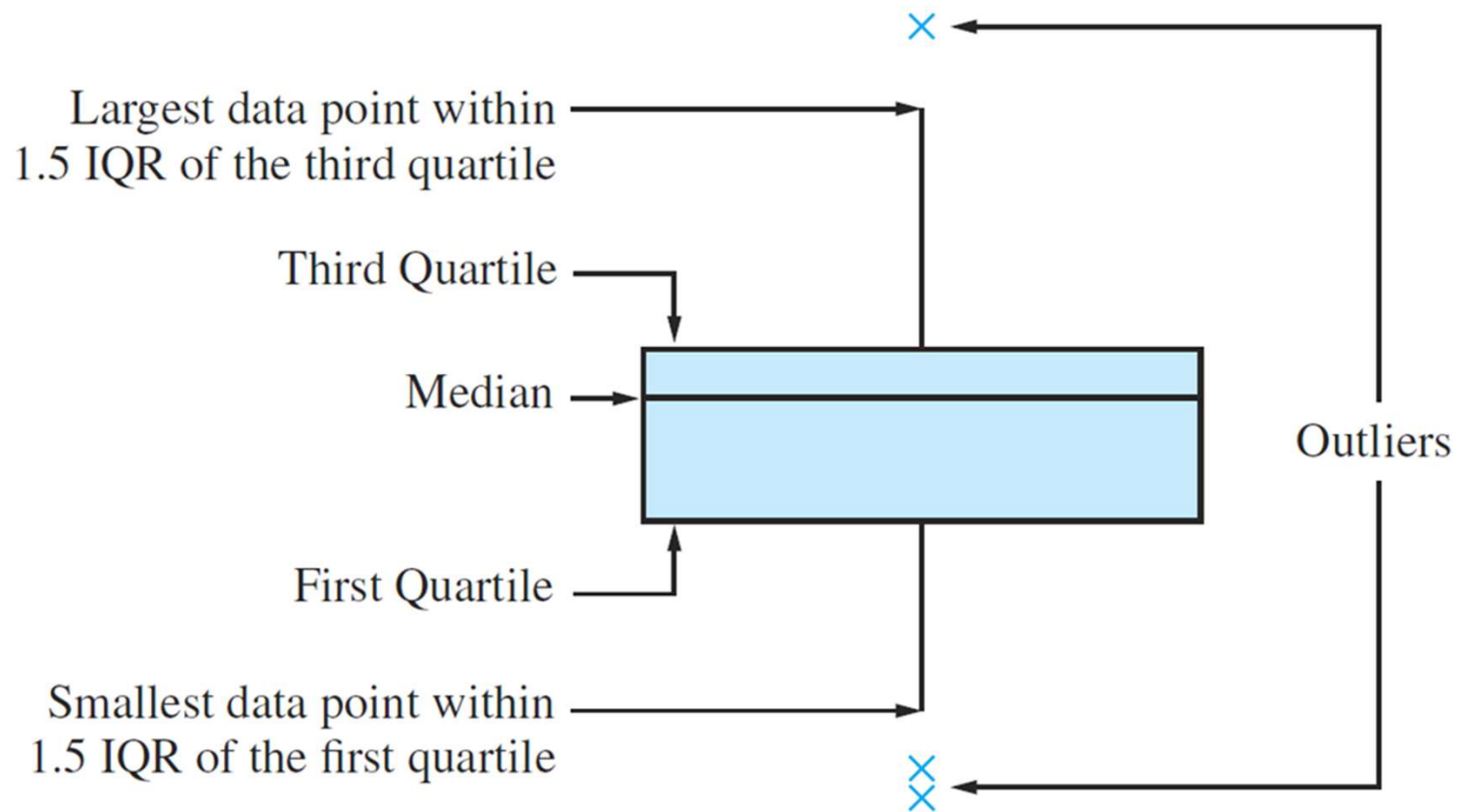
Friendship Paradox

Assumptions:

- The people with many friends constitute a smaller portion of the population.
- I am a randomly chosen person.
- I am more likely to be friends with a person who has many friends.



Anatomy of a Boxplot



Example 5

- Notice there are no outliers in these data.
- Looking at the four pieces of the boxplot, we can tell that the sample values are comparatively densely packed between the median and the third quartile.
- The lower whisker is a bit longer than the upper one, indicating that the data has a slightly longer lower tail than an upper tail.
- The distance between the first quartile and the median is greater than the distance between the median and the third quartile.
- This boxplot suggests that the data are skewed to the left.

Stem Leaf

4 259

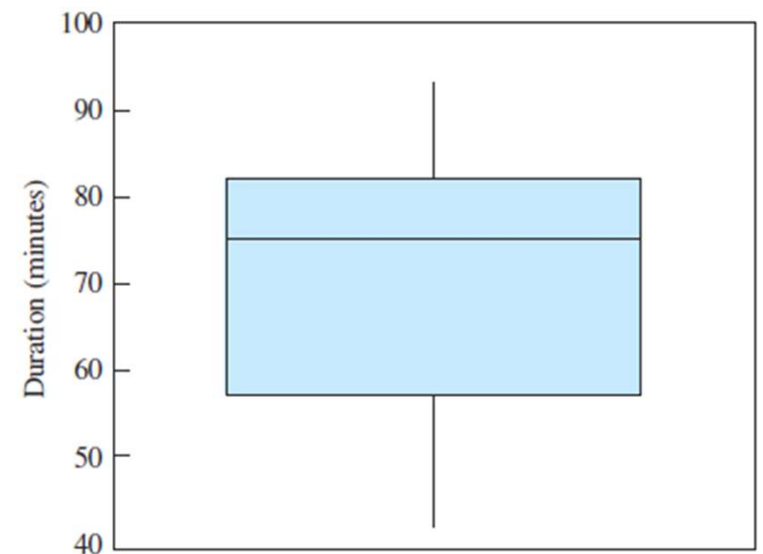
5 0111133556678

6 067789

7 01233455556666699

8 000012223344456668

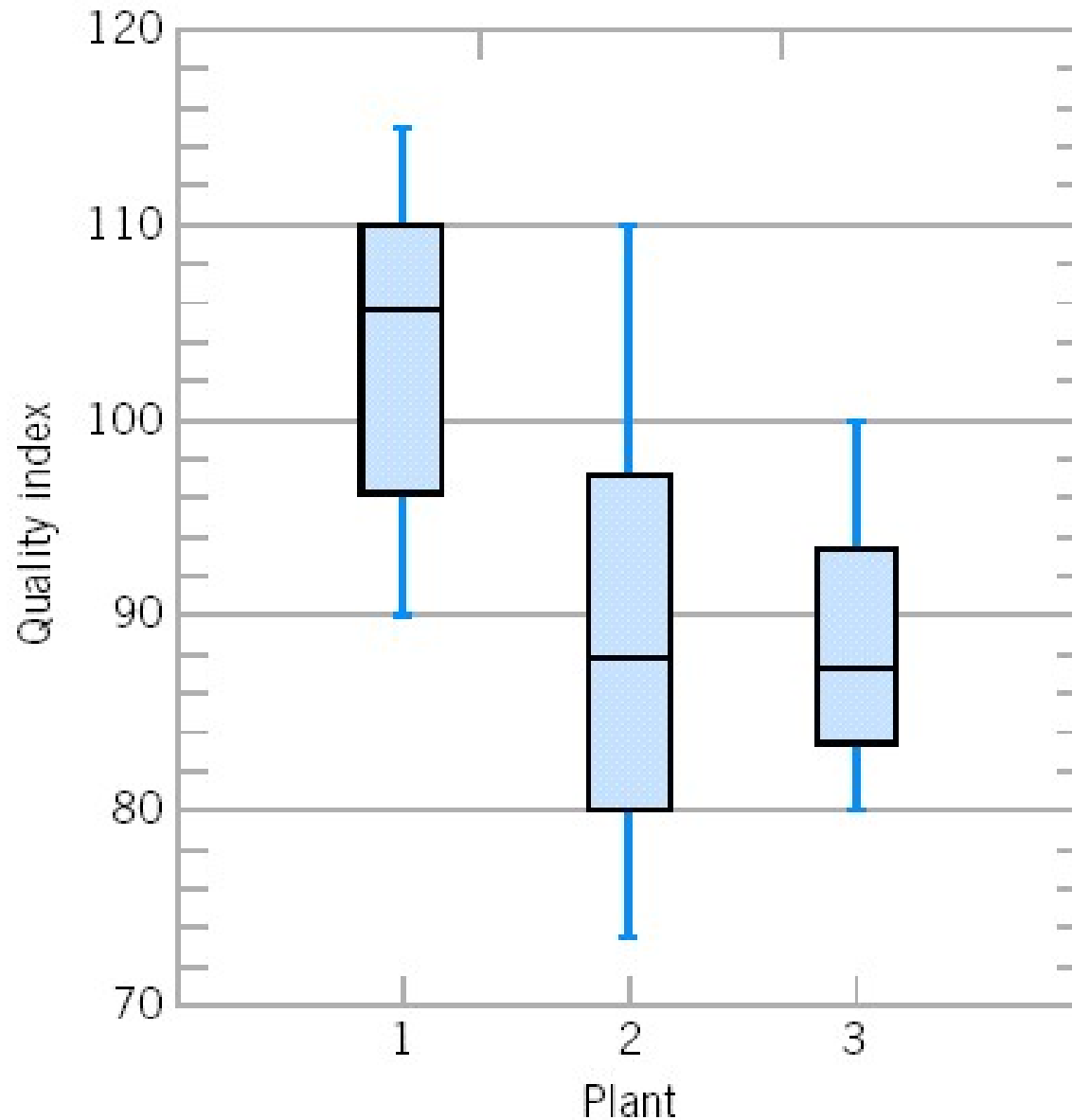
9 013



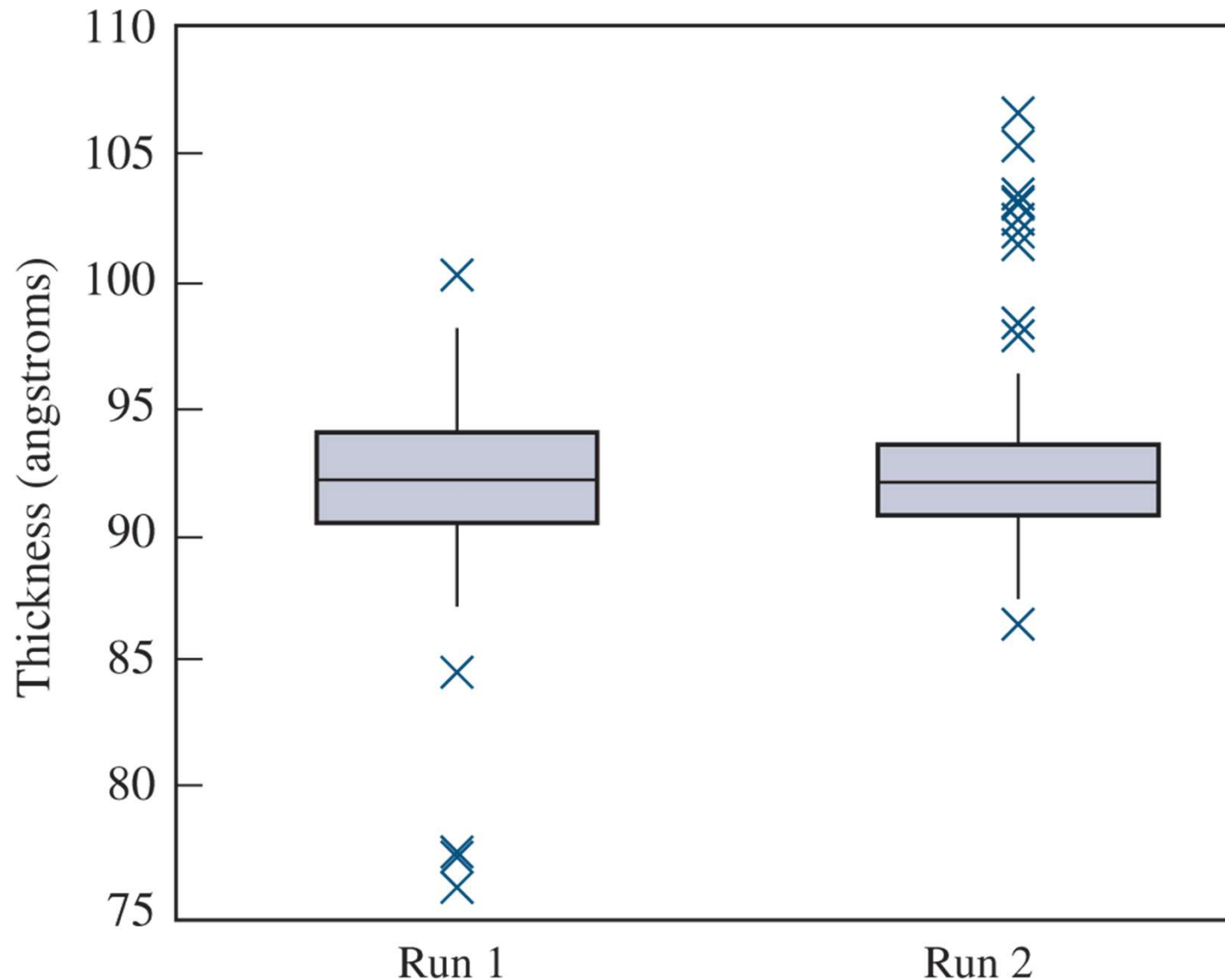
Comparative Boxplots

- Sometimes we want to compare more than one sample.
- We can place the boxplots of the two (or more) samples side-by-side.
- This will allow us to compare how the medians differ between samples, as well as the first and third quartile.
- It also tells us about the difference in spread between the two samples.

Comparative box plots of a quality index at three plants

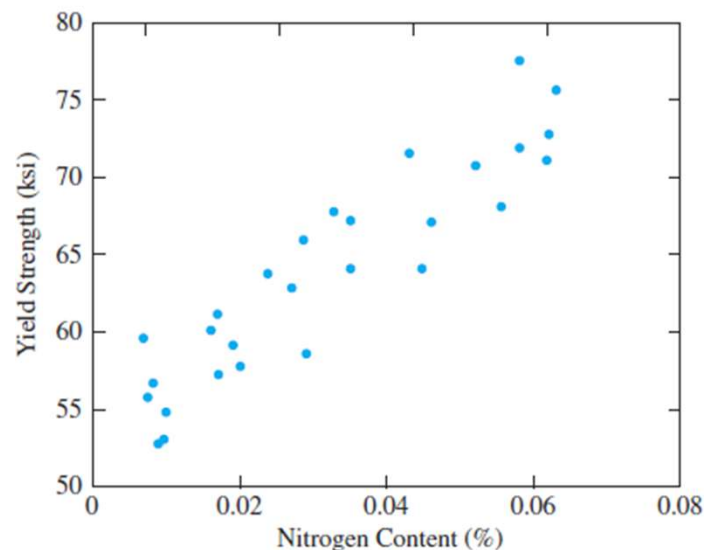


Comparative box plots of oxide layer thickness data



Scatterplot

- Data for which items consists of a pair of values is called **bivariate**.
- The graphical summary for bivariate data is a **scatterplot**.
- Display of a scatterplot:



Looking at Scatterplots

- If the dots in the scatterplot do not appear to have a general trend or pattern, then there is little evidence of a relationship between the two variables.
- If the dots on the scatterplot appear to roughly follow a straight line, then there is evidence of a linear relationship between the two variables.
- When a linear relationship exists, we may be able to use one variable to help predict the values of the other variable.

Summary of Chapter 1

- We discussed types of data.
- We looked at sampling methods, with an emphasis on simple random sampling (SRS).
- We learned about sample statistics.
- We examined graphical displays of data.